

Tikkavarapu Hema Sai

Coimbatore, India — +91 9059587397 — hemasairedy2006@gmail.com —
github.com/Hemasai23266 — LinkedIn

Professional Summary

Computer Science undergraduate with hands-on experience in software development, full-stack web development, machine learning, computer vision, and cloud computing. Proficient in Python, Java, JavaScript, React.js, Node.js, SQL, TensorFlow, OpenCV, and Git. Passionate about developing AI-powered applications, scalable software solutions, and solving real-world engineering problems through innovative technologies.

Technical Skills

Programming Languages: Python, Java, C, JavaScript, SQL

Web Technologies: React.js, Node.js, HTML5, CSS3, REST APIs, Responsive Web Design

Machine Learning & Computer Vision: TensorFlow, Keras, CNN, OpenCV, Image Processing, Data Augmentation, Model Evaluation

Cloud & Development Tools: AWS, Git, GitHub, VS Code, Jupyter Notebook, CI/CD Fundamentals

Core Concepts: Data Structures & Algorithms, Database Management Systems, Edge Computing, Problem Solving

Projects

Facial Expression Recognition Using CNN

Python, TensorFlow, Keras, OpenCV

- Developed a CNN-based deep learning model to classify human facial expressions into multiple emotion categories.
- Applied image preprocessing, histogram equalization, noise reduction, and data augmentation to improve model accuracy.
- Evaluated model performance using accuracy, confusion matrix, and F1-score for effective emotion classification.
- GitHub Repository: github.com/Hemasai23266/Facial-Expression-Recognition

Accessible Language Learning Platform

React.js, Node.js, REST APIs

- Developed a responsive learning platform for users with dyslexia, ADHD, and auditory processing limitations.
- Built interactive React.js interfaces with personalized learning modules and adaptive content delivery.
- Designed RESTful APIs for user authentication, progress tracking, and adaptive learning workflows.

Vehicle Near Collision Detection

Python, OpenCV, YOLOv5, Deep SORT

- Developed a computer vision system for real-time vehicle detection, tracking, and near-collision prediction.
- Integrated YOLOv5 object detection with Deep SORT tracking to assign persistent IDs across video frames.
- Implemented Time-to-Collision (TTC) estimation to identify potential collision risks for intelligent traffic monitoring.
- GitHub Repository: github.com/Hemasai23266/Vehicle-Near-Collision-Detection

Education

Amrita Vishwa Vidyapeetham

B.Tech. in Computer Science and Engineering

2023 – 2027

Coimbatore, India

Additional Information

Areas of Interest: Artificial Intelligence, Machine Learning, Computer Vision, Full-Stack Development, Cloud Computing

Activities: Hackathons, Team Coding Events, Cricket, Volleyball